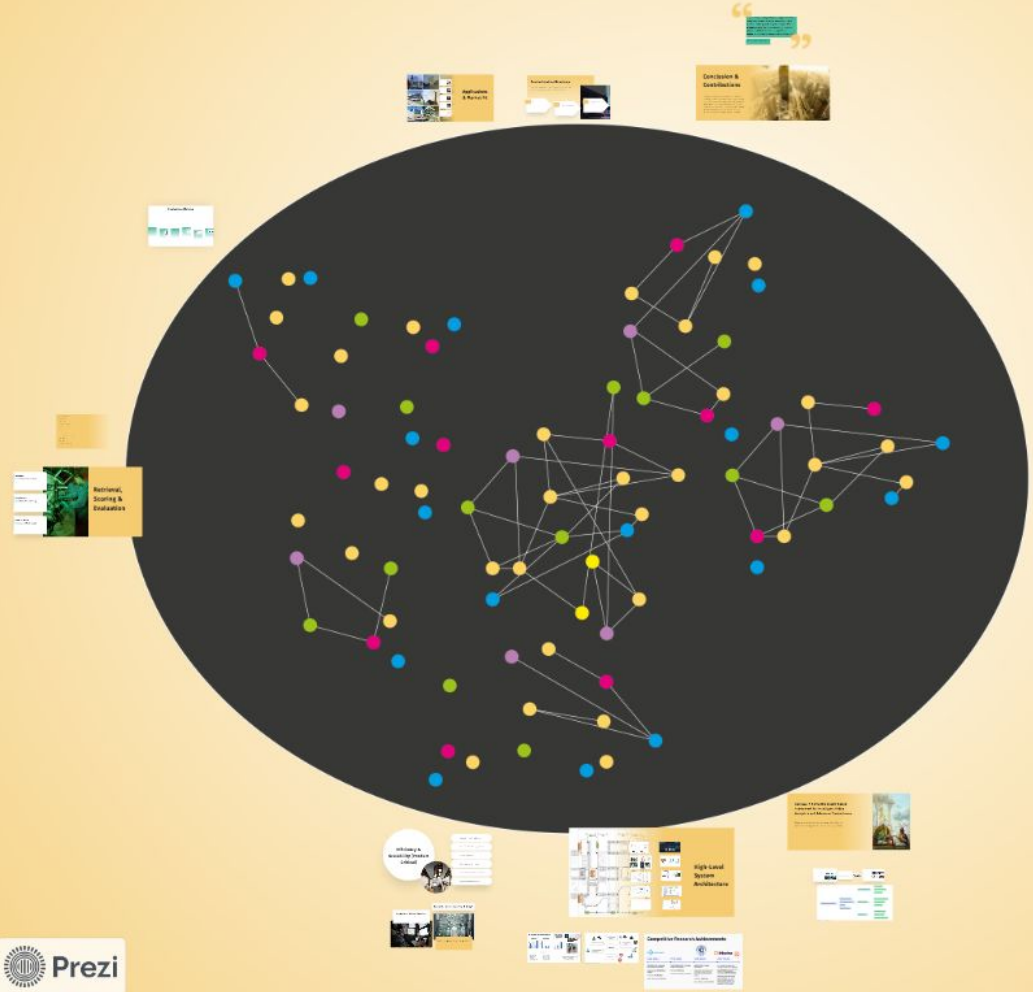


Sanjaya: A Semantic Graph-Based Framework for Intelligent Video Analytics and Advanced Surveillance

A framework transforming video data into structured, actionable insights for effective surveillance

Project Code: 5125

Prezi

Massive Video Data & Generation Limitations of Human Monitoring

- Surveillance systems generate massive volumes of video data in modern cities.
- Continuous human monitoring is impractical at scale.
- Post-incident analysis relies on manual frame-by-frame review, consuming tens of thousands of human hours.
- Existing systems offer frame-level detection and anomaly flags.
- They lack semantic and temporal understanding, creating a gap between video data and actionable intelligence.

Prezi

Literature Survey

- Conducted an in-depth literature survey of existing video analytics and surveillance systems.

30+ Research Articles

Technical Domain / Field	IEEE / ACM	arXiv / Open Access	Industry / Institutional	Total
Efficient Edge-Cloud & Compute	3	1	1	5
VLMs & Vision-Language Grounding	1	4	1	6
Reasoning, RAG & Agentic AI	0	5	0	5
Anomaly & Event Detection	5	1	1	7
KGs & Semantic Scene Analysis	2	2	1	5
Summarization & Keyframe Extraction	2	0	0	2
Totals	13	13	4	30

Prezi

- **Current Systems - Detection/Classification Based**

- Storing the data in static

- Detection, Recognition & Flagging using CV-ML

NO Context & Reasoning

NO Context & Reasoning

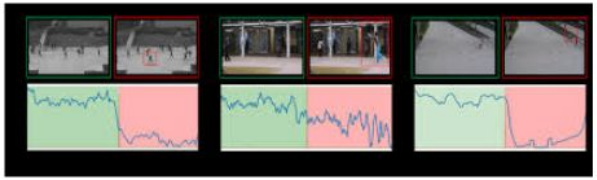
NO Explainability

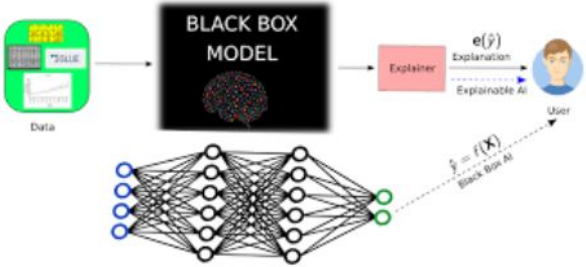
NO Query Capability

Prezi

Need for Actionable Intelligence

- **Detection ≠ Understanding**
- Organizations need systems that generate actionable intelligence from video data, not just store raw footage.
- Current systems often only collect video, which remains largely unused or **underutilized**.
- Pin-point **accurate insights** cannot be easily extracted from raw video footage.
- There is no efficient mechanism to **trace events, people, or activities** when required.
- Manual review of footage is time-consuming and inefficient.
- The gap is not detection accuracy — it is the **lack of structured, explainable understanding.**



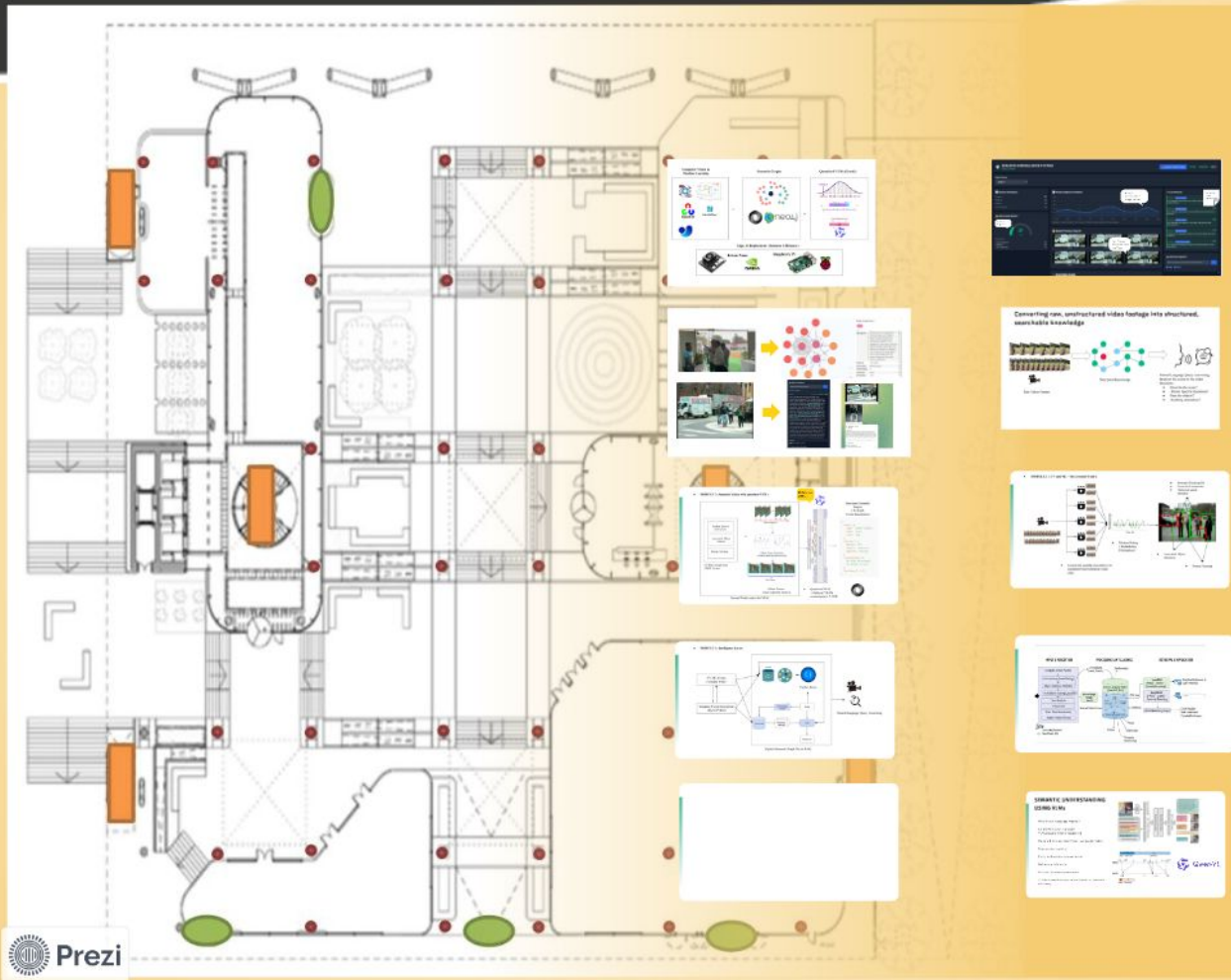


Prezi

Sanjaya: A Semantic Graph-Based Framework for Intelligent Video Analytics and Advanced Surveillance

Designed to transform surveillance video data into actionable intelligence for real-world applications.

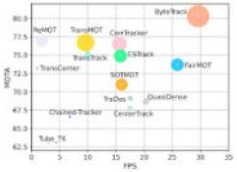
Prezi



High-Level System Architecture

Prezi

Computer Vision & Machine Learning



OpenCV

MediaPipe

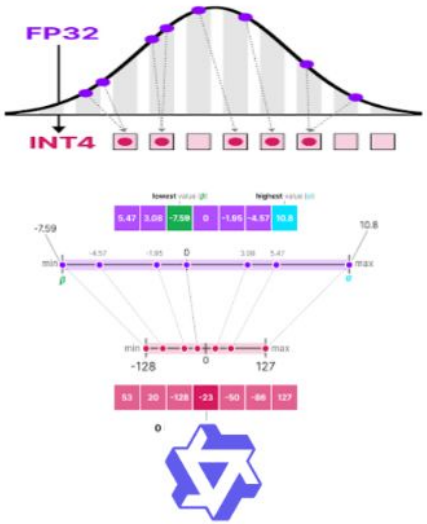
+

Semantic Graphs

neo4j

+

Quantized VLMs (GenAI)



Edge AI Deployment (Resource Efficiency)

Jetson Nano

NVIDIA

Raspberry Pi

Prezi

SANJAYA SURVEILLANCE SYSTEM

System: ONLINE

Upload Video/Images | CV: ON | Neo4j: ON | 50ms

Select Source:

seq01 ✓

Source Summary

Frames:	10
Persons:	4
Events:	5
VLM Analyses:	5

Risk Assessment

Risk Score Alignment

21

LOW

Risk Factors:

• Event Frequency	25
• Duration	25
• Zone Sensitivity	18

Frame Saliency Timeline

Activity Distribution in image frames

Red peaks = salient moments | Click to jump to frame | Currently showing: 10 frames analyzed

Salient Frames (Top-K)

Key Frames Extracted from the Video

Knowledge Graph

Live Events

Semantic Event Descriptions

05:30:00 ENTER SCENE
Four individuals are present in an outdoor urban s
street intersect...

05:30:00 ENTER SCENE LOW
Four individuals are present on a city street at 0.27s. One person is sitting on...

05:30:00 ENTER SCENE LOW
Four individuals are visible on a city street, with three of them standing and o...

05:30:00 ENTER SCENE LOW
A group of four individuals is present in an urban setting, with three people st...

05:30:00 POSTURE CHANGE LOW
Four people are entering the scene at a street intersection. Three individuals a...

Ask the System

What suspicious activities were detected?

☑ Graph ☑ Vector

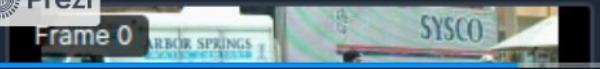
Prezi

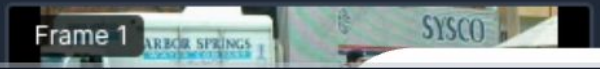
📈 Frame Saliency Timeline

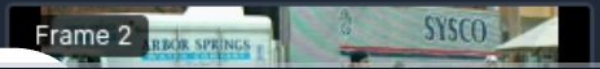
Activity Distribution in image frames

Red peaks = salient moments | Click to jump to frame | Currently showing: 10 frames analyzed

🖼️ Salient Frames (Top-K)







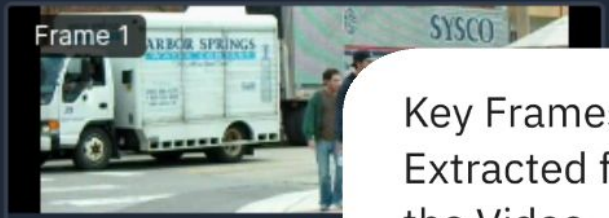
Prezi

0.0s 0.5s 1.0s 1.5s 2.0s 2.5s 3.0s 3.5s 4.0s 4.5s

Red peaks = salient moments | Click to jump to frame | Currently showing: 10 frames analyzed

Salient Frames (Top-K)

0.07s



0.17s

0.27s

Key Frames Extracted from the Video

0.30s

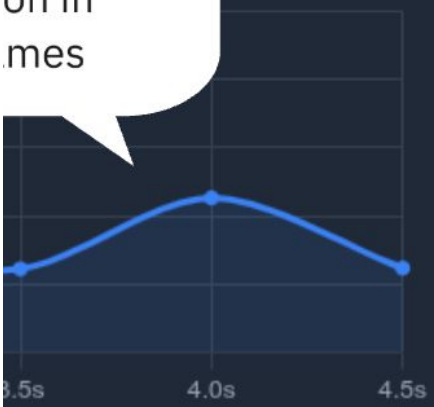
0.33s

0.37s

Knowledge Graph

Prezi

on in
mes



Prezi

⚡ Live Events

Semantic Event Descriptions

05:30:00 ENTER SCENE

Four individuals are present in an outdoor urban s
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05:30:00 ENTER SCENE LOW

Four individuals are present on a city street at 0.27s. One person is sitting on...

05:30:00 ENTER SCENE LOW

Four individuals are visible on a city street, with three of them standing and o...

05:30:00 ENTER SCENE LOW

A group of four individuals is present in an urban setting, with three people st...

05:30:00 POSTURE CHANGE LOW

Four people are entering the scene at a street intersection. Three individuals a...

📊 Source Summary

Frames:	10
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Events:	5
VLM Analyses:	5

⚠️ Risk Assessment

Risk Score Alignment

Risk Factors:

• Event Frequency	25
• Duration	25
• Zone Sensitivity	18

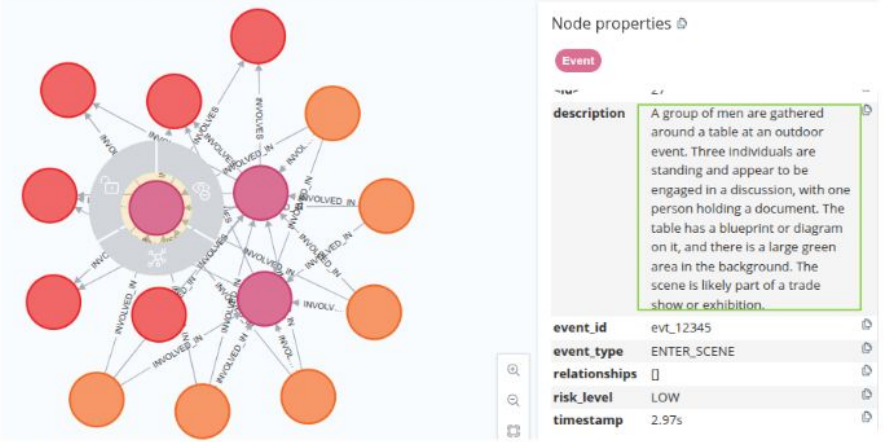
📈 Frame Saliency Time

Red peaks = salient moments | Click

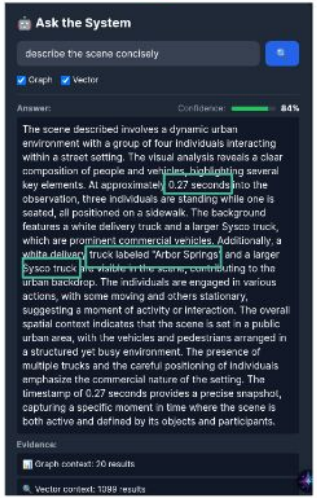
🖼️ Salient Frames (Top-

0.07s

Prezi


Node properties
Event
description
A group of men are gathered around a table at an outdoor event. Three individuals are standing and appear to be engaged in a discussion, with one person holding a document. The table has a blueprint or diagram on it, and there is a large green area in the background. The scene is likely part of a trade show or exhibition.
event_id
evt_12345
event_type
ENTER_SCENE
relationships
[]
risk_level
LOW
timestamp
2.97s
INVOLVED_IN

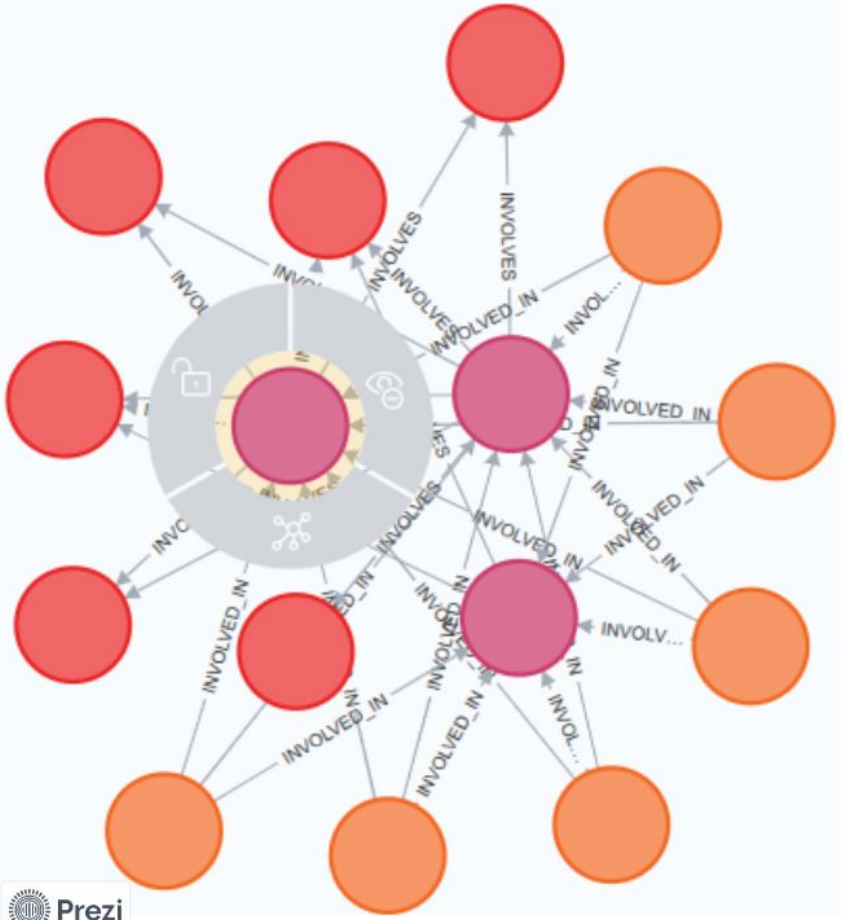
ARBOR SPRINGS
SYSCO


Ask the System
describe the scene concisely
Graph
Vector
Answer:
Confidence:
84%
The scene described involves a dynamic urban environment with a group of four individuals interacting within a street setting. The visual analysis reveals a clear composition of people and vehicles, highlighting several key elements. At approximately 0.27 seconds into the observation, three individuals are standing while one is seated, all positioned on a sidewalk. The background features a white delivery truck and a larger Sysco truck, which are prominent commercial vehicles. Additionally, a white delivery truck labeled "Arbor Springs" and a larger Sysco truck are visible in the scene, contributing to the urban backdrop. The individuals are engaged in various actions, with some moving and others stationary, suggesting a moment of activity or interaction. The overall spatial context indicates that the scene is set in a public urban area, with the vehicles and pedestrians arranged in a structured yet busy environment. The presence of multiple trucks and the careful positioning of individuals emphasize the commercial nature of the setting. The timestamp of 0.27 seconds provides a precise snapshot, capturing a specific moment in time where the scene is both active and defined by its objects and participants.
Evidence:
Graph context: 30 results
Vector context: 1099 results

SANJAYA
ENTER_SCENE @ 0.27s
Person crop from event
ENTERSCENE** @ 0.27s
Risk: LOW
Analysis:
At the 0.27-second mark, the scene captures a dynamic interaction involving four individuals at a street intersection. According to the semantic descriptions, the event features a group of people positioned in various postures: three individuals are standing, while one is sitting. This composition suggests a casual or transitional moment, possibly during a pause in movement. The visual focus is primarily on the individuals and their interactions with the surrounding environment.
The scene prom...
Entities: (Tracked): 2
RAG Sources: Graph + Vector

Prezi

Prezi



Node properties

Event

description	A group of men are gathered around a table at an outdoor event. Three individuals are standing and appear to be engaged in a discussion, with one person holding a document. The table has a blueprint or diagram on it, and there is a large green area in the background. The scene is likely part of a trade show or exhibition.
event_id	evt_12345
event_type	ENTER_SCENE
relationships	[]
risk_level	LOW
timestamp	2.97s

Prezi

ARBOR SPRINGS
WATER COMPANY
ARBOR SPRINGS
WATER COMPANY
SYSCO
MICHIGAN

Ask the System

describe the scene concisely

Graph Vector

Answer: Confidence: 84%

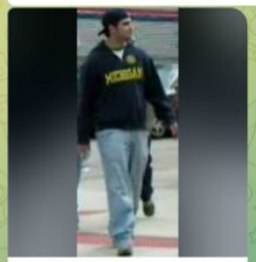
The scene described involves a dynamic urban environment with a group of four individuals interacting within a street setting. The visual analysis reveals a clear composition of people and vehicles, highlighting several key elements. At approximately 0.27 seconds into the observation, three individuals are standing while one is seated, all positioned on a sidewalk. The background features a white delivery truck and a larger Sysco truck, which are prominent commercial vehicles. Additionally, a white delivery truck labeled "Arbor Springs" and a larger Sysco truck are visible in the scene, contributing to the urban backdrop. The individuals are engaged in various actions, with some moving and others stationary, suggesting a moment of activity or interaction. The overall spatial context indicates that the scene is set in a public urban area, with the vehicles and pedestrians arranged in a structured yet busy environment. The presence of multiple trucks and the careful positioning of individuals emphasize the commercial nature of the setting. The timestamp of 0.27 seconds provides a precise snapshot, capturing a specific moment in time where the scene is both active and defined by its objects and participants.

Evidence:

Graph context: 20 results

SANJAYA
bot

ENTER_SCENE @ 0.27s 9:42 pm



Person crop from event 9:42 pm

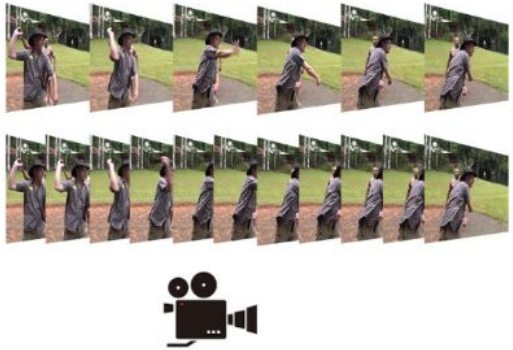
ENTER*SCENE** @ 0.27s
Risk: LOW

Analysis:
At the 0.27-second mark, the scene captures a dynamic interaction involving four individuals at a street intersection. According to the semantic descriptions, the event features a group of people positioned in various postures: three individuals are standing, while one is sitting. This composition suggests a casual or transitional moment, possibly during a pause in movement. The visual focus is primarily on the individuals and their interactions with the surrounding environment.

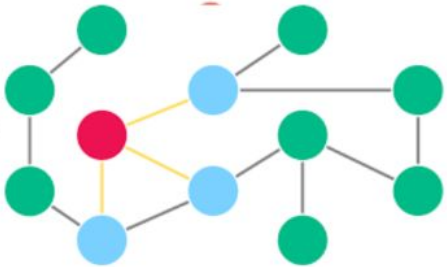
The scene promi...

Entities: {'trackid': 2}
RAG Sources: Graph + Vector

- **Converting raw, unstructured video footage into structured, searchable knowledge**



Raw Video Frames



Structured Knowledge

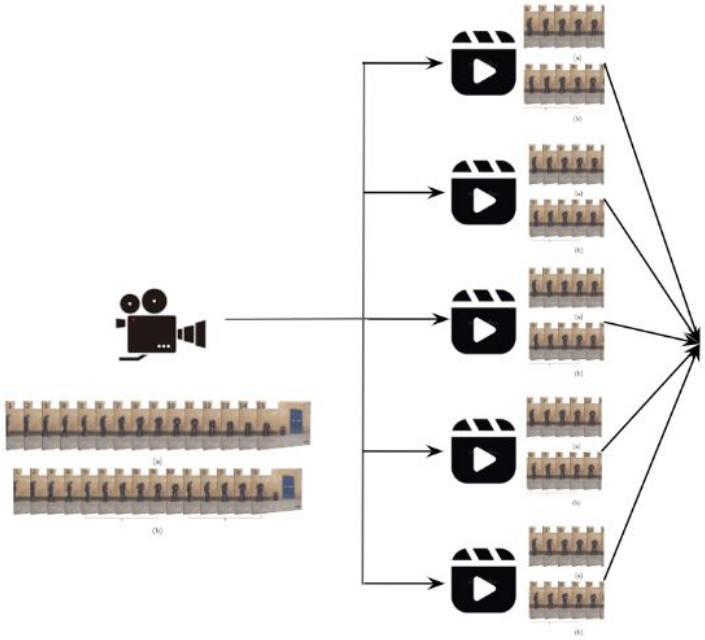
Natural Language Query Answering based on the events in the video

Questions:

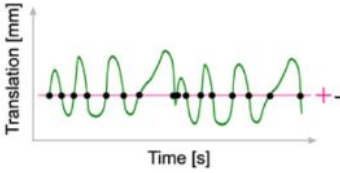
- Describe the scene?
- Person Specific Questions?
- State the objects?
- Anything anomalous?

Prezi

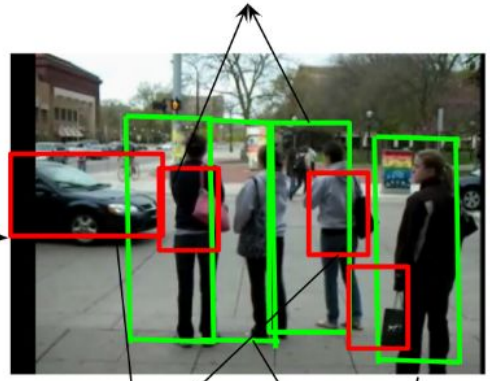
- **MODULE 1 (CV and ML - The Ground Truth)**



- Concurrent parallel execution over segmented short-duration video clips.



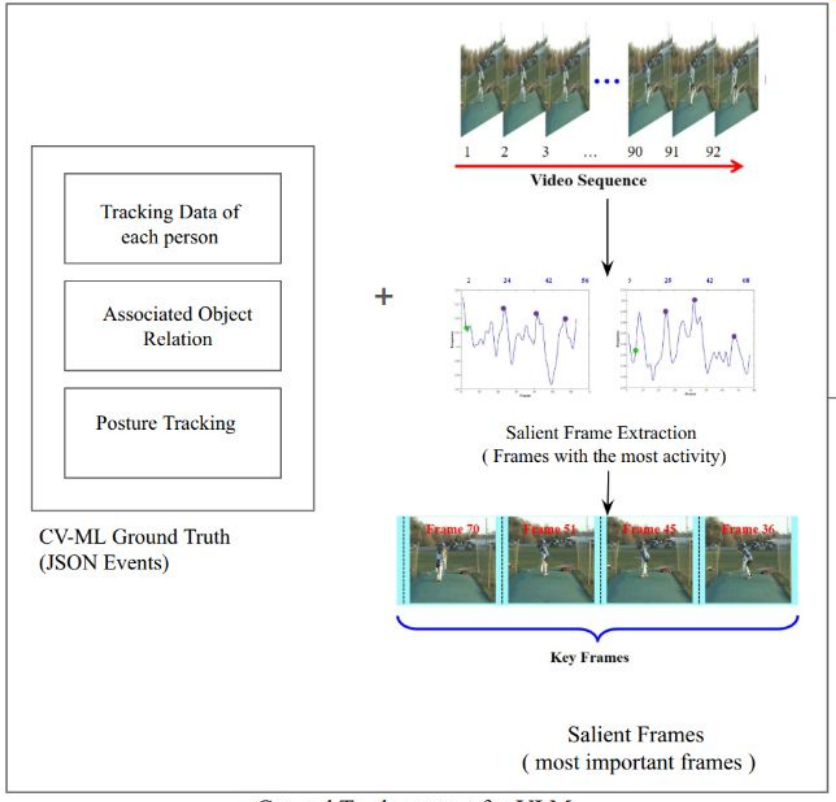
- Motion-Gating (Redundancy Elimination)



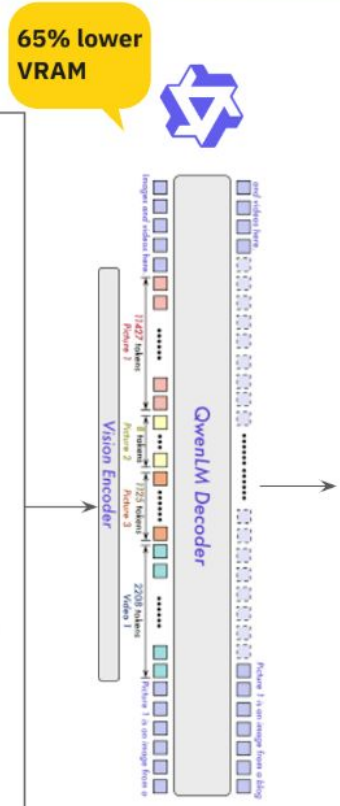
- Persistent Tracking IDs
- Event-level summaries
- Motion & speed estimates
- Associated Object Detection
- Posture Tracking

Prezi

- **MODULE 2 (Semantic Vision with quantised VLM)**



Ground Truth context for VLM



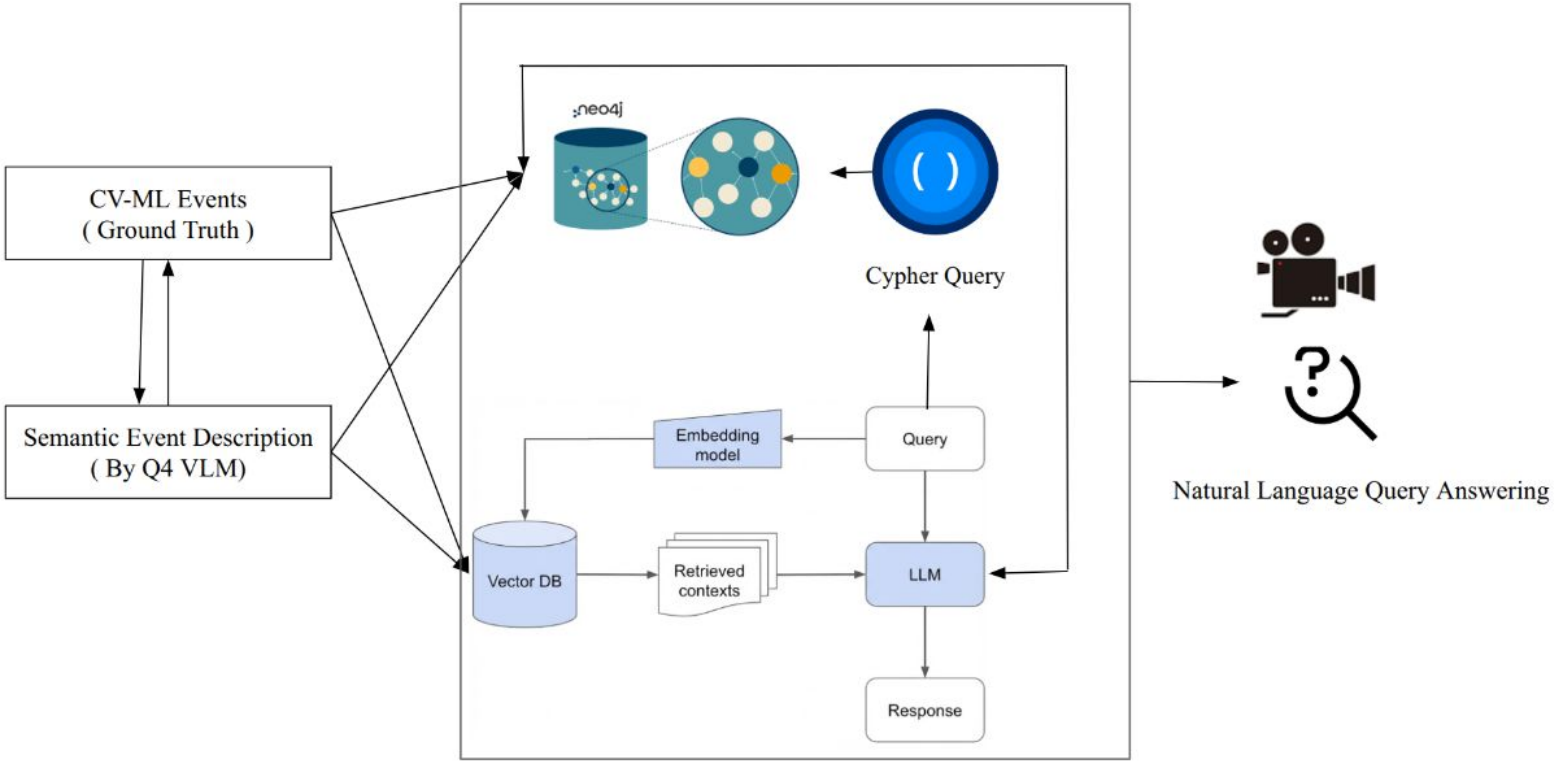
Structured Semantic Output (In Depth Event Description)

```
{
  "event": {
    "type": "OBJECT_PICKUP",
    "time": "2.0s",
    "risk": "LOW"
  },
  "entities": {
    "person": "P2",
    "object": "backpack",
    "posture": "walking"
  },
  "relationships": [
    "P2 PICKS_UP backpack",
    "P2 MOVING in Zone1"
  ],
  "confidence": 0.75
}
```

- Quantized VLM (Reduced VRAM consumption)- 3.5 GB

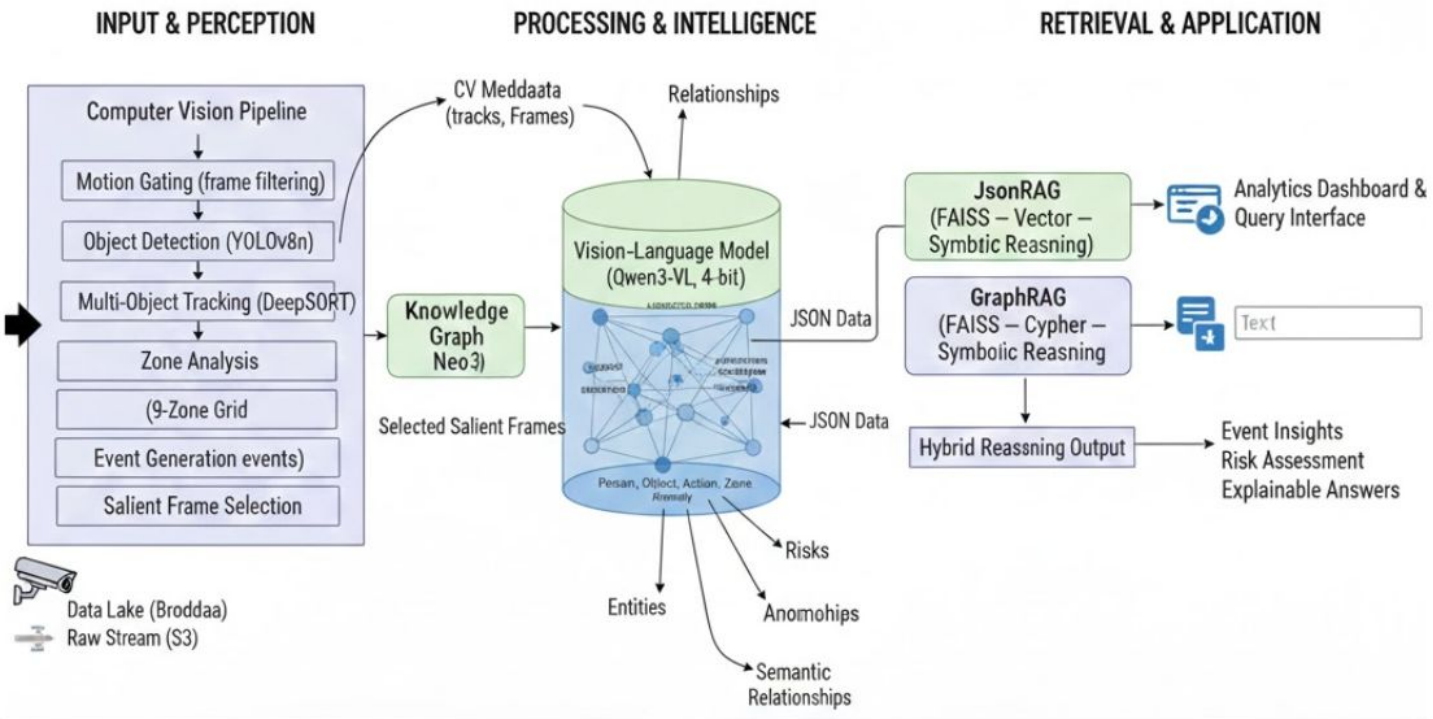
Prezi

- **MODULE 3 (Intelligence Layer)**



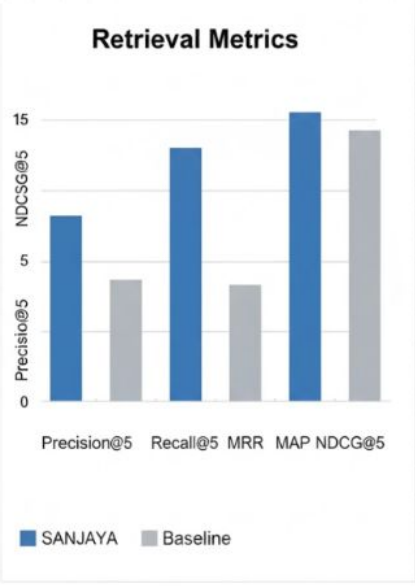
Hybrid Semantic Graph Vector RAG

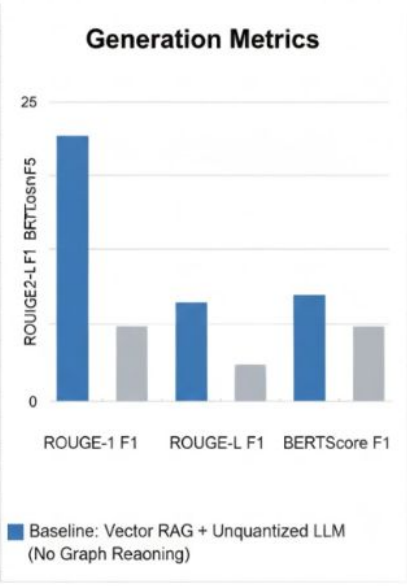
Prezi


INPUT & PERCEPTION
PROCESSING & INTELLIGENCE
RETRIEVAL & APPLICATION
Computer Vision Pipeline
Motion Gating (frame filtering)
Object Detection (YOLOv8n)
Multi-Object Tracking (DeepSORT)
Zone Analysis
(9-Zone Grid
Event Generation events)
Salient Frame Selection
Data Lake (Broddaa)
Raw Stream (S3)
CV Meddaata (tracks, Frames)
Relationships
Knowledge Graph Neo3)
Selected Salient Frames
Vision-Language Model (Qwen3-VL, 4-bit)
JSON Data
JSON Data
Pesan, Object, Action, Zone
Entities
Risks
Anomohips
Semantic Relationships
JsonRAG (FAISS – Vector – Symbtic Reasning)
GraphRAG (FAISS – Cypher – Symbolic Reasning
Hybrid Reassning Output
Analytics Dashboard & Query Interface
Text
Event Insights
Risk Assessment
Explainable Answers

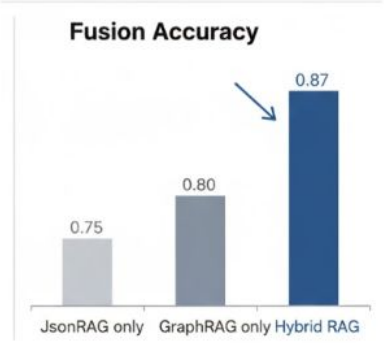
Prezi

Evaluation and Benchmarking





16% retrieval accuracy with Hybrid RAG

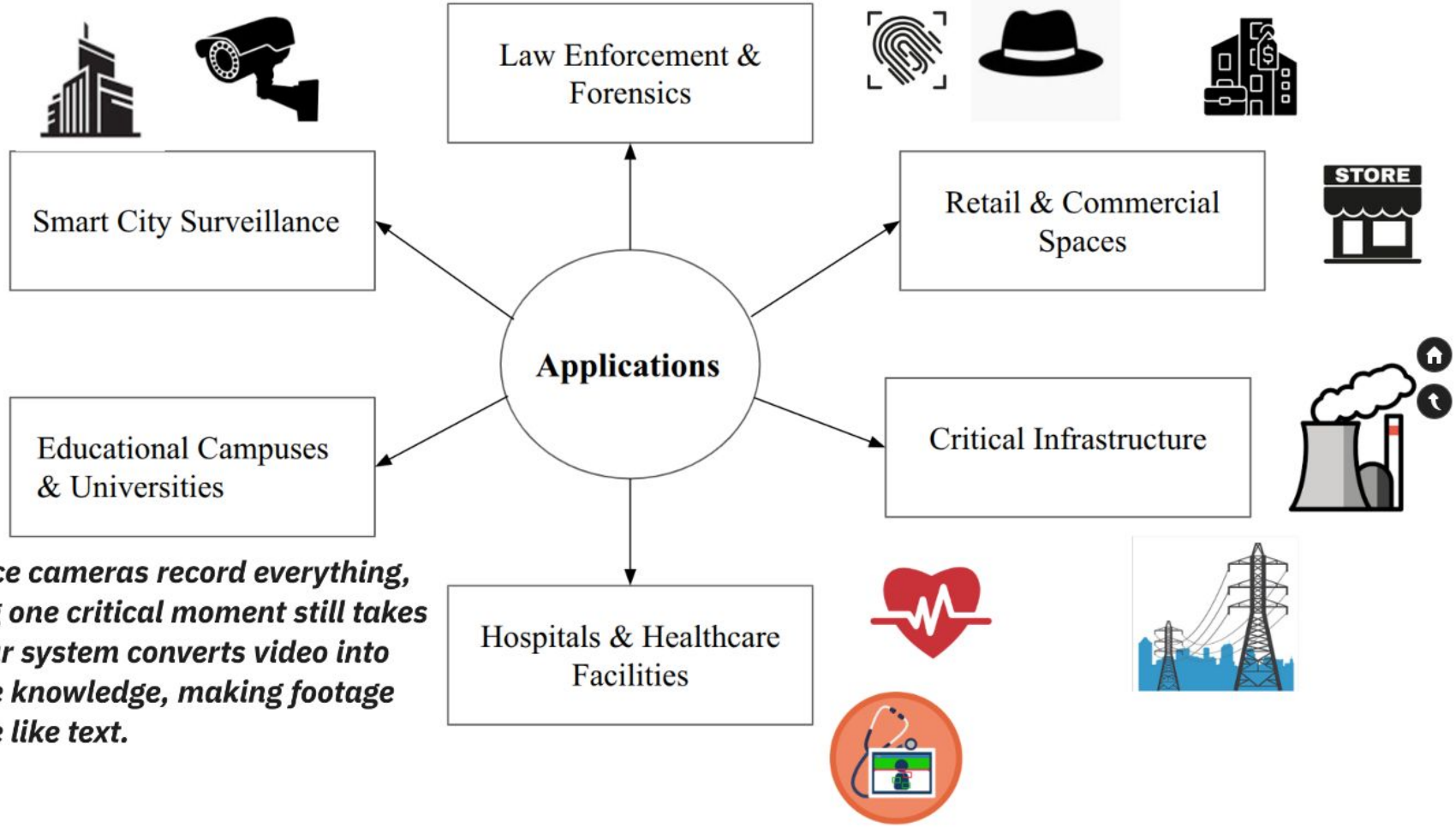


- Precision@5: **0.87**
- Recall@5: **0.82**
- MRR: **0.91**
- **12–18% improvement** over baseline

- ROUGE-1 F1: **0.72**
- ROUGE-L F1: **0.68**
- BERTScore F1: **0.82**
- Outputs show **high semantic consistency with actual events**

Benchmarked on Dataset by Stanford University & University of Michigan

Prezi



Surveillance cameras record everything, but finding one critical moment still takes hours — our system converts video into searchable knowledge, making footage searchable like text.

Prezi

Competitive Research Achievements

DEC 2024

PULZION 2024 – Research Presentation Competiton

Organized by **ACM Student Chapter**

Outcome: **First Position**

Level: City-Level Competition

FEB 2025

TECHTHESIS 2025 – Technical Project Competition

Outcome: **First Prize**

Level: Inter-College Competition

APR 2025

ASPIRE-2K25 – Project Competition

Organized in association with tComputer Society of India (CSI), and Society for Data Science

Outcome: **Third Prize**

Level: State-Level Competition

OCT 2025

The research project has attracted interest from **iMocha** and **Dwarpaal**, international AI startups with valuations exceeding **USD 100 million**.

Standing paid internship opportunities have been communicated following technical discussions and evaluation of the project.

Prezi